

Debaleen Das Spandan

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- Springfield, MO
- LinkedIn
- Portfolio

Skills

- Programming Languages: Kotlin, Python, C, C++, Java, JavaScript
- Frameworks & Libraries: PyTorch, OpenCV, Pandas, Numpy, Android SDK, REST APIs, WebRTC, Authentication (SSO/OAuth)
- DevOps/Tools: Git, GitHub Actions, Jira, Linux, Gradle, Docker
- AI & Computer Vision: Convolutional Neural Networks (CNN), Multimodal Sensing, Scene Graphs

Professional Experience

Software Engineer

Digital Monitoring Products, Springfield, MO

Aug 2024 - Present

- AlarmVision:** Developed C-based camera firmware enabling integration between security panels and camera hardware systems.
 - Reduced firmware size from ~150 MB to under 10 MB by transitioning from a Go-based implementation to C, contributing to improved interoperability with external hardware vendors.
- Real-time Camera Streaming (WebRTC):** Designed and implemented a modular WebRTC video streaming component enabling live camera monitoring within the Android application.
 - Integrated with multiple signaling backends including Unreal Media Server, AWS Kinesis Video Streams, and custom third-party media servers to support diverse device ecosystems.
 - Reduced live video startup time from ~5–10 seconds to under 3 seconds and improved end-to-end latency to under 500 ms.
- App Architecture & Feature Integration:** Improved application maintainability and scalability for an 84K+ user base by refactoring legacy components and introducing modular design patterns aligned with Jetpack architecture guidelines.
 - Reduced technical debt by ~30% through removal of deprecated dependencies and expansion of unit test coverage, improving release stability.
 - Implemented secure SSO authentication workflows enabling a more streamlined and reliable customer login experience.
 - Supported ongoing application reliability by resolving customer-facing issues and enabling compatibility with newly released security panel hardware.

Graduate Research Assistant

Missouri State University, Springfield, MO

Aug 2022 - Jul 2024

- ProxeGraph:** Led development of a computer-vision pipeline for proxemics-driven human–device interaction monitoring to enable accessible non-verbal smart-home automation workflows.
 - Designed real-time scene understanding modules for object detection and spatial distance modeling using Python, OpenCV, YOLOv8 (Ultralytics), and NumPy.
- Metamorphic Testing Framework for Voice-Driven Smart-Home Systems:** Developed test cases and testing strategies to evaluate robustness of context recognition models under varied input transformations.
- Student Attention Monitoring System:** Built a real-time computer-vision prototype to classify student engagement states using pose key-point extraction and deep learning–based modeling with MediaPipe.
- Technical Training & Mentorship:** Developed and delivered hands-on workshops for educators on practical cybersecurity vulnerabilities including denial-of-service attacks, Wi-Fi exploitation, and IoT security risks and mentored undergraduate students in debugging, programming fundamentals, and applied computer science concepts.

Research Engineer

National Institute of Technology, Durgapur, India

Jun 2022 - Jul 2022

- Multimodal Accessibility Interface:** Developed a prototype system integrating interactive frontend components with machine-learning-based perception modules to enable IoT device control through diverse input modalities. Led a team of 4 on end-to-end development and research experimentation.
- Tech Stack: Python, React.js, Node.js, TensorFlow.js, Mediapipe.

Programmer Analyst

Cognizant, India

Jan 2021 - Jun 2022

- Automated configuration and workflow processes using Python and AWS Lambda to streamline data-handling pipelines and reduce operational overhead.
- Implemented enhanced client-side logging and monitoring features using TypeScript that improved debugging efficiency for large web applications in enterprise finance and supply-chain domains.

Research Intern	Feb 2020 - Aug 2020
National Institute of Technology , Durgapur, India	
• BuStop: Built data preprocessing and visualization modules for a sensor-driven public transportation prediction system using Python-based analytics tools. • IndoAir: Developed an Android application embedding pre-trained models to estimate indoor PM2.5 and CO₂ levels from environmental inputs and visualize classroom-level AQI through interactive map views with actionable improvement suggestions. • Tech Stack: Python, JavaScript, Pandas, NumPy.	

Education	
MS in Computer Science, Missouri State University	Aug 2022 - Jul 2024
Springfield, Missouri	
• Thesis: Scene Understanding and Spatial Analysis using Hall's Proxemics zone in smart homes • Software Engineering Certificate	
B.Tech in Computer Science, Bengal College of Engineering & Technology	Aug 2017 - Jul 2021
Durgapur, India	
• Relevant Coursework: Software Engineering, Operating Systems, Database Management Systems, Algorithms, Computer Networks, Data Structures, Object Oriented Programming	

Publications	
1. To Hack or Not to Hack? Connecting Cybersecurity, the IoT, and Smart Environments to K12 Learning Experiences	2025
Shusmoy Chowdhury, <i>Debaleen Das Spandan</i> , Tara Bonebrake, Zachary Frantsen, Ajay Katangur. Foundations of Computer Science and Frontiers in Education: Computer Science and Computer Engineering, Springer, 2025. DOI: 10.1007/978-3-031-85930-4_28	
2. ProxeGraph: Scene Graph Generation Utilizing Proxemics for Smart Homes	2024
<i>Debaleen Das Spandan</i> , Razib Iqbal. IEEE MIPR 2024. DOI: 10.1109/MIPR62202.2024.00024	
3. Scene Understanding and Spatial Analysis Using Scene Graph Enhanced by Hall's Proxemics Zones in Smart Homes	2024
<i>Debaleen Das Spandan</i> . Master's Thesis, Missouri State University, 2024. BearWorks: 3989	
4. Metamorphic Testing for Investigation of Context Recognition from Smart Home Voice Commands	2024
Fahima Hasan Athina, Jeniya Sultana, <i>Debaleen Das Spandan</i> , Razib Iqbal. IEEE CIoT 2024. DOI: 10.1109/CIoT63799.2024.10757078	
5. Exploiting Multi-modal Contextual Sensing for City-bus's Stay Location Characterization: Towards Sub-60 Seconds Accurate Arrival Time Prediction	2023
Ratna Mandal, Prasenjit Karmakar, Soumyajit Chatterjee, <i>Debaleen Das Spandan</i> , Shouvit Pradhan, Sujoy Saha, Sandip Chakraborty, Subrata Nandi. ACM Transactions on Internet of Things, vol. 4, no. 1, Feb 2023, Article 1. DOI: 10.1145/3549548	
6. Can I Take Class in That Classroom Now? In the Perspective of Air Quality	2023
Praveen Kumar Sharma, Debjit Chatterjee, <i>Debaleen Das Spandan</i> , Sujoy Saha. Proc. 24th International Conference on Distributed Computing and Networking (ICDCN 2023). DOI: 10.1145/3571306.3571442	

Awards	
• IEEE TCMC Student Travel Support — IEEE	Aug 2024
• Graduate Student Travel Fund — Graduate College, Missouri State University	Jul 2024
• Outstanding Poster Presentation — 31st Annual Frank Einhellig Graduate Interdisciplinary Forum, Missouri State University	Apr 2024
• CS-CNAS Thesis Research Funding — Graduate College, Missouri State University	Feb 2023